

TULLN LANGENLEBARN, Austria

WMO: 110300

Lat: 48.3239N

Lon: 16.1181E

Elev: 580

StdP: 14.39

Time Zone: 1.00 (EUC)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	12.8	17.8	7.1	8.1	14.7	12.0	10.4	19.9	31.8	38.7	28.5	40.3	5.1	250	0.543

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	20.3	90.0	71.1	86.4	69.6	83.4	68.0	72.5	86.7	70.7	84.0	69.1	81.2	9.2	130

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
67.7	104.1	79.8	66.1	98.2	77.4	64.5	93.0	75.3	36.4	86.8	34.9	84.3	33.5	81.1	79.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
26.4	23.0	20.2	6.1	95.3	7.5	3.8	0.7	98.0	-3.7	100.2	-7.9	102.3	-13.3	105.1		
			5.8	75.5	7.3	1.8	0.5	76.8	-3.7	77.9	-7.9	78.9	-13.2	80.3		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	51.3	32.1	35.7	42.7	51.7	59.7	66.4	69.5	68.8	60.2	50.9	42.4	34.5	(6)
(7)		DBStd	14.89	8.20	8.52	7.40	7.21	6.48	6.58	5.96	5.78	6.06	6.99	6.99	7.65	(7)
(8)		HDD50	2079	558	402	244	66	4	0	0	0	2	76	244	483	(8)
(9)		HDD65	5466	1021	819	691	403	189	62	24	26	167	438	679	947	(9)
(10)		CDD50	2551	1	2	18	116	305	492	605	584	308	104	15	1	(10)
(11)		CDD65	463	0	0	0	2	25	103	164	145	23	1	0	0	(11)
(12)		CDH74	4572	0	0	0	38	272	1045	1625	1385	201	6	0	0	(12)
(13)		CDH80	1518	0	0	0	4	43	332	608	495	36	0	0	0	(13)
(14)	Wind	WSAvg	8.3	9.4	9.4	9.4	8.7	8.6	7.5	8.0	7.2	7.4	7.3	8.5	8.3	(14)
(15)	Precipitation	PrecAvg	24.6	1.1	1.1	1.7	1.7	2.8	3.2	3.3	2.8	2.4	1.6	1.6	1.4	(15)
(16)		PrecMax	33.2	3.0	2.7	5.0	3.5	6.1	6.8	9.9	7.0	7.5	3.5	3.0	2.2	(16)
(17)		PrecMin	16.7	0.1	0.0	0.3	0.0	0.7	1.3	0.4	0.7	0.6	0.1	0.0	0.4	(17)
(18)		PrecStd	4.3	0.7	0.8	1.1	1.0	1.4	1.4	1.7	1.7	1.7	0.9	0.7	0.6	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	55.8	59.3	68.5	78.7	84.5	91.8	95.0	95.3	84.1	73.5	64.8	54.6	(19)
(20)			MCWB	48.3	48.6	53.7	60.4	66.6	72.2	71.3	70.6	67.1	61.6	55.7	48.6	(20)
(21)		2%	DB	51.1	55.0	64.1	73.5	80.3	87.5	90.2	89.6	79.2	68.4	59.4	50.8	(21)
(22)			MCWB	45.5	47.4	51.5	58.5	65.1	70.9	71.1	70.7	65.4	59.2	53.3	46.3	(22)
(23)		5%	DB	46.7	51.7	59.1	69.7	76.9	84.0	86.5	85.4	75.3	65.1	55.8	47.1	(23)
(24)			MCWB	42.4	45.2	49.2	56.2	63.0	69.0	69.6	69.1	63.6	57.7	51.0	43.4	(24)
(25)		10%	DB	43.1	48.1	55.0	65.9	73.3	80.5	83.1	81.7	71.8	61.8	52.3	44.2	(25)
(26)			MCWB	39.8	43.0	47.1	54.1	61.0	66.9	68.1	67.4	61.8	56.1	48.4	41.3	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	49.6	50.2	54.9	61.8	69.0	74.4	74.7	74.1	69.2	62.8	56.6	49.5	(27)
(28)			MCDB	55.0	57.1	66.5	75.7	81.3	87.7	89.6	88.9	81.5	70.2	62.6	53.6	(28)
(29)		2%	WB	45.9	47.7	52.2	59.0	66.1	72.0	72.5	71.7	66.7	60.5	54.2	46.6	(29)
(30)			MCDB	50.6	54.0	61.6	71.7	77.7	85.2	87.2	85.9	77.0	67.0	59.3	50.6	(30)
(31)		5%	WB	42.7	45.5	50.2	56.9	63.8	69.9	70.6	69.9	64.7	58.6	51.3	43.8	(31)
(32)			MCDB	46.3	51.1	57.5	68.3	74.4	81.9	84.0	83.3	73.6	64.0	55.7	46.6	(32)
(33)		10%	WB	40.0	43.2	47.9	54.8	61.9	67.7	69.0	68.2	62.7	56.6	48.8	41.4	(33)
(34)			MCDB	42.8	47.9	54.3	64.1	71.6	78.5	80.8	79.9	70.2	61.1	52.0	43.8	(34)
(35)	Mean Daily Temperature Range	MDBR	9.2	12.8	16.2	19.5	19.2	20.0	20.3	20.5	17.4	14.6	10.1	8.8	(35)	
(36)		5% DB	MCDBR	14.0	19.1	25.5	27.2	25.6	26.2	27.2	27.7	23.8	19.6	15.9	13.5	(36)
(37)			MCWBR	10.1	12.5	15.0	14.4	12.7	11.6	10.8	11.3	11.5	11.7	10.5	9.8	(37)
(38)		5% WB	MCDBR	13.9	17.7	23.2	25.3	23.6	24.2	24.7	25.1	22.5	17.7	15.1	12.6	(38)
(39)			MCWBR	10.4	12.0	14.2	13.8	12.1	11.4	10.7	11.0	11.4	11.1	10.3	9.6	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.318	0.342	0.383	0.428	0.427	0.432	0.439	0.438	0.409	0.377	0.349	0.317	(40)	
(41)		taud	2.374	2.330	2.253	2.160	2.207	2.232	2.215	2.236	2.283	2.344	2.378	2.403	(41)	
(42)		Ebn at Noon	235	255	261	260	265	264	260	255	250	238	220	220	(42)	
(43)		Edh at Noon	22	29	36	44	43	43	43	40	35	28	22	19	(43)	
(44)	All-Sky Solar Radiation	RadAvg	312	579	937	1397	1635	1785	1721	1537	1099	672	345	246	(44)	
(45)		RadStd	42	76	122	150	181	142	164	151	138	78	35	28	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)		
(46)	Station Only		+1.17	N/A	N/A	N/A	+0.95	+1.21	N/A	N/A	N/A
(47)	Regional (91 neighbors)		+0.97	N/A	N/A	+1.44	+0.74	+0.61	-218	-274	+135

Nomenclature: See separate page